Generating the energy-constrained completion by completely positive maps

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Abstract

We show that every element of the specified energy-constrained completion is a difference of two completely positive elements. A uniform decomposition in finite dimensions passes to the completion through a summable positive series. An exact infimum formula for the norm follows.

Repository target. This manuscript concerns AIM problem 928 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

Answer. The answer to the precise completion question is affirmative. The key step is a dimension-independent completely positive decomposition on a dense space of maps supported on finite input and output subspaces. An absolutely convergent series then puts both components in the specified closed cone, rather than merely in the larger cone of maps with finite energy-constrained norm.

Theorem 1.1. *Let $\mathcal{H}_A, \mathcal{H}_B$ be separable complex Hilbert spaces, and suppose that $G \geq 0$ on $\mathcal{H}_A$ has compact resolvent, minimum eigenvalue zero, and eigenvalues tending to infinity. Fix $E > 0$. Let Y be the real vector space of Hermiticity-preserving maps on trace class with finite ordinary diamond norm, let X_G be its completion for $q(T) = \|T\|^G_{\diamond,E}$, and let C_G be the closure in X_G of its completely positive cone. Then*

$$X_G = C_G - C_G.$$

For every $x \in X_G$ and $\eta > 0$, the components may be chosen so that $x = a - b$ and $q(a + b) \leq 2q(x) + \eta$.

The quantitative constant two is sufficient for the theorem. A refinement below replaces it by one, with the same arbitrarily small additive error. No attainment of a minimal decomposition is asserted.

1.1 Subnormalized inputs and finite-dimensional approximation

Choose a unit vector $g \in \ker G$. If $\sigma \geq 0$ is a subnormalized input–ancilla state with $\operatorname{Tr}\sigma \leq 1$ and $\operatorname{Tr}((G \otimes I)\sigma) \leq E$, then

$$\|(T \otimes \mathrm{id})(\sigma)\|_1 \leq q(T). \tag{928.1}$$

Indeed, in an enlarged ancillary space add $(1 - \operatorname{Tr}\sigma)|g\rangle\langle g|$ in an orthogonal ancillary flag. The resulting state is normalized and still has energy at most E. Its output is a direct sum, whose trace norm is at least that of the output on σ. An infinite-dimensional separable ancilla accommodates the extra flag isometrically. This argument requires no positivity or trace-preservation assumption on T.

Let P_n be increasing finite-rank spectral projections of G, containing the ground eigenspace, with $P_n \to I$ strongly. Choose $g_n \to \infty$ so that $G \geq g_n(I - P_n)$. For every normalized state ρ of input energy at most E, the gentle projection bound gives

$$\|\rho - (P_n \otimes I)\rho(P_n \otimes I)\|_1 \leq 2\sqrt{\operatorname{Tr}((I - P_n)\rho_A)} \leq 2\sqrt{E/g_n}.$$

Consequently, for $T \in Y$,

$$q(T - T \circ \operatorname{Ad}_{P_n}) \leq 2\|T\|_\diamond \sqrt{E/g_n} \longrightarrow 0, \qquad \operatorname{Ad}_V(\rho) = V\rho V^*. \tag{928.2}$$

Fix n, put $d = \operatorname{rank} P_n$, and restrict the input to $P_n\mathcal{H}_A \cong \mathbb{C}^d$. The resulting map has a trace-class Choi operator

$$J_n = \sum_{i,j=1}^{d} T(|e_i\rangle\langle e_j|) \otimes |e_i\rangle\langle e_j| \in \mathcal{T}_1(\mathcal{H}_B \otimes \mathbb{C}^d).$$

If $Q_m \to I$ strongly are finite-rank output projections, then

$$(Q_m \otimes I_d)J_n(Q_m \otimes I_d) \longrightarrow J_n \quad \text{in trace norm.}$$

For any Hermiticity-preserving map R with finite-dimensional input,

$$\|R\|_\diamond \leq \|J_R\|_1. \tag{928.3}$$

To see this, purify an input state. Up to an ancillary isometry its output has the form $(I \otimes \sqrt{\rho})J_R(I \otimes \sqrt{\rho})$, with $\|\sqrt{\rho}\| \leq 1$; a transpose can be absorbed into the input convention. Partial trace contracts the trace norm of Hermitian operators, so mixed inputs cause no difficulty. Thus the maps

$$T_{n,m}(\rho) = Q_m T(P_n \rho P_n) Q_m$$

approximate $T \circ \operatorname{Ad}_{P_n}$ in ordinary diamond norm, and hence in q, as $m \to \infty$.

Let D be the real linear space of maps with such finite input and output support. Two of them can be placed in common larger subspaces, so D is indeed linear. Equations (928.2)–(928.3) show that D is dense in Y for q, and therefore dense in X_G.

1.2 A uniform positive decomposition on the dense space

Take $T \in D$, with input support $P = P_n$. On the finite-dimensional space $P\mathcal{H}_A$, write $G_P = G|_{P\mathcal{H}_A}$ and set

$$W = (I + G_P/E)^{-1/2}, \qquad S = T \circ \operatorname{Ad}_W .$$

Here W^{-1} is bounded because the input space is finite dimensional. For any normalized state ω on this input space and an ancilla, $\sigma = (W \otimes I)\omega(W \otimes I)$ has trace at most one and energy at most E, since $WG_PW = E(I - W^2) \leq EI$. By (928.1),

$$\|S\|_\diamond \leq q(T). \tag{928.4}$$

A finite-dimensional Hermiticity-preserving map admits completely positive maps S_+, S_- such that

$$S = S_+ - S_-, \qquad \|S_+ + S_-\|_\diamond = \|S\|_\diamond. \tag{928.5}$$

For clarity, this is the Hermitian form of the diamond-norm semidefinite program [1]. The precise Choi-form dual is given in [3, Theorem 6 and Section 3.2, pp. 10–12]. For a Hermitian Choi matrix, average a feasible two-block dual with its block swap. The two diagonal blocks then agree, and conjugation by the two-by-two Hadamard matrix gives the constraints below. If J is the Hermitian Choi matrix, its dual minimizes t subject to $Z \geq J$, $Z \geq -J$, and $\operatorname{Tr}_B Z \leq tI$. Strong duality and attainment hold in finite dimensions. The Choi matrices $(Z + J)/2$ and $(Z - J)/2$ give (928.5); a completely positive map has diamond norm equal to the operator norm of its trace functional.

Extend $S_\pm$ by zero outside $P\mathcal{H}_A$ and define ordinary bounded completely positive maps

$$T_\pm = S_\pm \circ \mathrm{Ad}_{W^{-1}P}\,.$$

They have finite input and output support, and $T = T_+ - T_-$. For any full input–ancilla state ρ of energy at most E, put

$$\sigma = (P \otimes I)\rho(P \otimes I), \qquad \tau = (W^{-1} \otimes I)\sigma(W^{-1} \otimes I).$$

All operators in this formula act on the finite input subspace, and

$$\operatorname{Tr}\tau = \operatorname{Tr}\sigma + E^{-1}\operatorname{Tr}((G_P \otimes I)\sigma) \leq 2.$$

Moreover, $(T_+ + T_-) \otimes \mathrm{id}(\rho) = (S_+ + S_-) \otimes \mathrm{id}(\tau)$. Therefore

$$q(T_+ + T_-) \leq 2\|S_+ + S_-\|_\diamond \leq 2q(T). \tag{928.6}$$

Complete positivity also gives $q(T_\pm) \leq q(T_+ + T_-)$, because the positive output of either component has trace at most the trace of their sum. This establishes the required uniform decomposition without using an infinite-dimensional normal-map decomposition or an unbounded inverse.

1.3 An absolutely convergent series proves exact cone membership

Let $x \in X_G$ and $\epsilon > 0$. Choose $x_k \in D$ with $q(x - x_k) \leq \epsilon 2^{-k}$ for $k \geq 1$. Put $d_1 = x_1$ and $d_k = x_k - x_{k-1}$ for $k \geq 2$. Then

$$\sum_{k\geq 1} q(d_k) \leq q(x) + 2\sum_{k\geq 1} q(x - x_k) \leq q(x) + 2\epsilon. \tag{928.7}$$

For each k, write $d_k = a_k - b_k$ using (928.6). The series $a = \sum_k a_k$ and $b = \sum_k b_k$ converge in X_G, since $q(a_k), q(b_k) \leq 2q(d_k)$. Every finite partial sum is an ordinary bounded completely positive map. By the definition of C_G, both limits belong to C_G. Finally,

$$a - b = \sum_{k\geq 1} d_k = \lim_{k\to\infty} x_k = x, \qquad q(a + b) \leq 2q(x) + 4\epsilon.$$

Taking $\epsilon < \eta/4$ proves Theorem 1.1.

1.4 Refinement of the quantitative bound

In fact, for $T \in D$ there is a decomposition with $q(T_+ + T_-) = q(T)$. Here is a direct finite-dimensional derivation. Use an eigenbasis of G_P and let $J = J_T$. Purification and trace-norm duality identify $q(T)$ with

$$\begin{aligned} &\text{maximize} && \operatorname{Tr}(JX), \\ &\text{subject to} && \rho \geq 0, \quad \operatorname{Tr}\rho = 1, \quad \operatorname{Tr}(G_P\rho) \leq E, \\ &&& -I_B \otimes \rho \leq X \leq I_B \otimes \rho. \end{aligned} \tag{928.8}$$

The input transpose in the Choi convention leaves G_P unchanged. The finite-input norm equals the full-space norm of T: any compressed full-space input is subnormalized and can be padded within the input subspace, as in (928.1).

Let $A, B \geq 0$ multiply the two matrix inequalities in (928.8), and let $\lambda \geq 0$ and $\mu \in \mathbb{R}$ multiply the energy and trace constraints. Supremizing the Lagrangian forces $A - B = J$ and $\mathrm{Tr}_B(A + B) \leq \mu I + \lambda G_P$. Thus, with $Z = A + B$, the dual is

$$\begin{aligned} \text{minimize} \quad & \mu + \lambda E, \\ \text{subject to} \quad & Z \geq J, \quad Z \geq -J, \\ & \mathrm{Tr}_B\, Z \leq \mu I + \lambda G_P, \qquad \lambda \geq 0. \end{aligned} \tag{928.9}$$

There is a positive-definite density matrix of energy strictly below E, obtained by concentrating sufficiently strongly on a ground vector. With $X = 0$ this gives strict primal feasibility. Hence the dual optimum is attained and equals $q(T)$.

The positive Choi matrices $(Z \pm J)/2$ give $T_\pm$. Their sum has energy-constrained norm at most $\mu + \lambda E = q(T)$; the reverse inequality follows by applying the triangle inequality to the two positive outputs. Extending outside the input subspace is harmless: the dual inequality on a ground vector implies $\mu \geq 0$, so it remains valid on the omitted spectral subspace. Substituting this improved decomposition into (928.7) proves that, for every $\eta > 0$, one can arrange

$$x = a - b, \quad a, b \in C_G, \qquad q(a + b) \leq q(x) + \eta.$$

Corollary 1.2 (Exact infimum formula)**.** *For every $x \in X_G$,*

$$q(x) = \inf\{q(a + b) : a, b \in C_G,\ x = a - b\}.$$

The formula concerns an infimum; it does not assert that an optimal pair exists.

Proof. For ordinary bounded completely positive maps A, B, positivity of the two outputs on every admissible input gives

$$\|(A - B) \otimes \mathrm{id}(\rho)\|_1 \leq \mathrm{Tr}((A + B) \otimes \mathrm{id}(\rho)).$$

Taking the supremum yields $q(A - B) \leq q(A + B)$. Given $a, b \in C_G$, choose ordinary bounded completely positive maps $A_n \to a$ and $B_n \to b$ in q. Continuity of the norm passes this inequality to the limit, giving $q(a - b) \leq q(a + b)$. This proves that the displayed infimum is at least $q(x)$. The decompositions with $q(a + b) \leq q(x) + \eta$ established above, for arbitrary $\eta > 0$, prove the converse inequality. □

Relation to the cited question. Shirokov's conjecture in Section 5 of [2] asks for components in the completed cone $\mathfrak{F}_G^0$. The decomposition following Lemma 2.19 in [4] produces components in the larger finite energy-norm cone. The series above addresses precisely the extra closure requirement: its two components are norm limits of ordinary bounded completely positive maps. This establishes the catalogue formulation [5] without identifying those two cones.

References

[1] J. Watrous, *Semidefinite programs for completely bounded norms*, Theory Comput. **5** (2009), 217–238. doi:10.4086/toc.2009.v005a011.

[2] M. E. Shirokov, *On completion of the cone of completely positive linear maps with respect to the energy-constrained diamond norm*, Lobachevskii J. Math. **40** (2019), 1549–1568. doi:10.1134/S1995080219100238. arXiv:1810.10922, Section 5.

[3] J. Watrous, *Simpler semidefinite programs for completely bounded norms*, arXiv:1207.5726v2, 2 August 2012, Theorem 6, p. 10, and Section 3.2, pp. 11–12.

[4] L. van Luijk, *Energy-Limited Quantum Dynamics*, Comm. Math. Phys. **406** (2025), article 120. doi:10.1007/s00220-025-05282-w. Lemma 2.19 and footnote 11.

[5] MColbrook/AIM, *AIM 928: Does the closed completely positive cone generate the energy-constrained completion?*, repository statement at revision `5d7e26e8444e`. https://github.com/MColbrook/AIM/blob/5d7e26e8444eb98de91988454564e45e011107c4/problems/928-does-the-closed-completely-positive-cone-generate-the-energy-constrained-completion.md.